RESULTS: CORRECTNESS & STRAGGLER TIMING

Aneesh Naresh | Team 31 | One Perlmutter node, four A100 GPUs

540
measured algorithm runs; all passed

0
maximum absolute error vs. NCCL

2
physical slow-GPU placements tested

Generated program speedup relative to NCCL AllReduce

Median of paired NCCL / generated latency ratios. Above 1.00x is faster; below 1.00x is slower.

Buffer / GPU	Slow GPU	0 ms delay	1 ms delay	5 ms delay
12 KiB	0	0.337x	0.970x	0.994x
12 KiB	3	0.315x	0.970x	0.994x
12 MiB	0	0.355x	0.806x	0.947x
12 MiB	3	0.385x	0.827x	0.954x
48 MiB	0	0.328x	0.661x	0.879x
48 MiB	3	0.323x	0.660x	0.878x

Generated code matched NCCL in every measured run. Speedup ranged from 0.315x to 0.994x across 18 cases; 0 cases had a median ratio above 1.00x.

Method. Signed inputs vary by GPU and element; every run starts from fresh buffers. NCCL is checked against a CPU sum, and all four outputs are checked after every run. Identical nominal delays are injected on the chosen physical GPU. Timing includes delay, ReduceScatter, communication and synchronization; setup and validation are excluded.

Scope. Five warmups and ten measured samples per algorithm/case in the default sweep. Host-observed latency includes launch overhead; clock-based delays are nominal. Small timing differences need repeated trials. Two-node / eight-GPU evaluation remains future work.

Evidence: retained raw logs and summary.csv | NCCL, handwritten and generated paths tested | November 7, 2026